

PCB Design and Layout Tools

In this month's EFY-CD, find some free, easy-to-use tools that let you create professional-quality PCBs

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There are three fundamental tasks that allow you to prepare a printed circuit board (PCB) for prototype and manufacture. First, the board outline must be created for the form factor of the design. Second is part placement, which involves configuring land patterns (or footprints) of various design devices on the board. The third fundamental task in board design is to route net connections between various parts. The routing process turns various net connections into copper traces. Copper traces connect parts in the physical prototype with current-carrying connections.

The net acts as a design guide indicating that two pins must be connected, while the copper trace is the actual physical connection which will be made as part of the PCB.

There are several software available that help you perform these tasks. Given below are a few of them.

DesignSpark PCB



DesignSpark PCB is a powerful and easy-to-use free PCB design tool. It gives all design engineers access to powerful, professional-standard software at no cost.

What's Special?

A comprehensive standard parts library is supplied, which can be amended and copied if and when required. Sophisticated new part creation wizards make it easy to design new parts from scratch or by using standard symbols and footprints. DesignSpark

PCB is intuitive and easy to learn and use. Free from practical constraints on board size, pin counts, layers and output types, it can be used for schematic capture, PCB layout and generating manufacturing files.

Distributor: Being free, this software can be downloaded directly from its website using an activation code. The activation code can be requested free of charge by registering your details at www.designspark.com/pcb-activation

*Home page: www.designspark.com/pcb
Download URL: <http://designspark.eu/downloads/released/programs/Setup.exe>
Platform Supported: Windows*

EAGLE 5.11.0



The latest 5.11.0 version of EAGLE is a powerful graphics tool for designing PCB layouts and schematics.

What's Special?

This easy-to-use software consists of three main modules: schematic editor, layout editor and autorouter. Schematic, layout and component editors have identical user interfaces. The layout editor has maximum drawing area of 1.6×1.6 m² (64×64 inch²) with resolution of 1/10,000 mm (0.1 micron). It supports up to 16 signal layers and comes with a full set of part libraries.

*Distributor: The software can be purchased online from website www.cadsoft.de/cgi-bin/order-us. There is a freeware version also, which can be downloaded from the website. Home page: www.cadsoftusa.com
Download URL: www.cadsoftusa.com/download.htm
Platform Supported: Linux, Mac, Windows*

ExpressPCB



ExpressPCB is a feature-rich PCB software for beginners.

What's Special?

ExpressPCB consists of two applications, one for drawing schematics and the other for PCB layout. It lets you complete your layout in a few simple steps. Same is the case with ExpressSCH. These both have the same user interface which makes it easy for the users.

Distributor: The software can be freely downloaded from the website.

*Home page: www.expresspcb.com
Download URL: www.expresspcb.com/expresspcb.htm/Download.htm
Platform Supported: Windows*

FreePCB



FreePCB is a free and Open Source PCB editor that is easy to learn and use, yet capable of professional-quality work.

What's Special?

FreePCB supports 1 to 16 copper layers and board size up to 152.4×152.4 cm² (60×60 inch²). It can import and export PADS-PCB netlists and has inbuilt design rule checker.

Distributor: Being free and Open Source, this software can be downloaded freely from the website of the software.

*Home page: www.freepcb.com
Download URL: www.freepcb.com/downloads.htm
Platform Supported: Windows*

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